US pool for a one-time restorative Parkinson's therapy: class-level demand

ClearPoint Neuro (CLPT) working paper, 24 Oct 2026. Scenario analysis, not guidance. Tags: A = published data, C = calculation, S = judgment (★), P = instruction from David May, D = document. Sources are listed at the end; every number traces to pd_pool_model.xlsx (sheet and cells shown under each table). Counts in tables are in thousands of patients.

Summary

- **Question.** How many US patients a year would have a one-time restorative PD procedure at class level if its label looked like DBS's? Not split across the four therapies and before ClearPoint's capacity. Disease-modifying use is excluded (section 7).
- **Funnel.** Medically eligible ⊃ referred ⊃ willing ⊃ covered ⊃ treated. Eligibility is an explicit scenario input. The only calibrated number is the downstream product referral × willing × covered (P), set so that modelled DBS under age 75 matches the 2019 volume (3.4k a year) [A10, S5].
- **Central eligibility: 8% of US PD (point prevalence) ★**, equal to 16.5% of incident patients becoming eligible at some point. At 8%, the implied P is 0.30. That matches an independent judgment of the stages (referral 55% × willing 60% × covered 90% = 0.30).
- 5% needs P = 0.48, which puts every stage near the top of its range.
- 15% (Aspen's 150k) needs P = 0.16, near the bottom.
- This fits David's view: Aspen is optimistic, and 10% is not double the most likely figure.
- **Class demand does not depend on the eligibility level once P is calibrated to DBS.** Eligible × P is pinned by DBS volume, so the four cases give the same demand. They differ only in the P they imply and how plausible it is. For eligibility to raise demand, the new class would need a better downstream funnel than DBS, which is outside scope.
- **Demand.** If DBS keeps its volume (bound A): 1.7k in the launch year (FY2029), 1.1k in FY2032, 2.0k in FY2040. If the therapy replaces DBS (bound B): 5.1k, 4.5k and 5.4k. At $20k per procedure, FY2040 is $39M–107M.
- **Main drivers (tornado, FY2040)**:
- The DBS volume used for calibration: 3.0k–4.5k gives bound B 4.7k–7.1k.
- The yearly draw-down of the backlog after launch: bound B 4.8k–5.8k.
- Eligibility, launch year, age cap, mortality and duration floor each move FY2040 demand by 0.1k or less.

1. Label and stage definitions

Figure 1. Funnel stages and what feeds each one

Stage	Definition in this model	Input	Tag
1 Medically eligible	Levodopa-responsive idiopathic PD, ≥4 yrs, motor complications not adequately controlled (recent-onset included), no dementia, fit, age <75	Scenario 5 / 8 / 10 / 15% of PD	A [B1], S3 ★, S8 ★
2 Referred / evaluated	Seen at a DBS-capable centre for assessment	Judgment 30–80% (comparison only)	S9 ★
3 Willing	Accepts surgery after evaluation	Judgment 40–85%; published 35–43% is consent to referral	A11, A12, A21, S10 ★
4 Covered	Payer approves	Judgment 80–95%	S4 ★
2×3×4 product P	Lifetime share of eligible patients who are treated	Calibrated: DBS under 75 ÷ eligible inflow, FY2019	C
5 Treated	DBS before launch; one-time therapy after launch	Flow model	C

Source: pd_pool_model.xlsx, Inputs and Scenarios.

- **The label does not require patients to be refractory.** Medtronic's label covers levodopa-responsive PD of at least 4 years that is not adequately controlled, including motor complications of recent onset (4 months to 3 years). It has no age cap [B1]. EARLYSTIM showed benefit in patients with early complications (age ≤60, H&Y <3, complications ≤3 years) [B7].
- **Age and prior DBS.** An upper age of 75 is a practice judgment (★ S3). All four restorative trials in ClearPoint's partner set exclude patients who have had DBS [D1–D4], so DBS recipients leave the pool permanently.
- **What each willingness figure measures.** The 35% [A11] and 43.2% [A12] figures are willingness among patients flagged as candidates, before referral. Of patients who complete a full evaluation, 58.7% go on to implant at one US centre [A21].

2. Medical eligibility: point prevalence and lifetime fraction

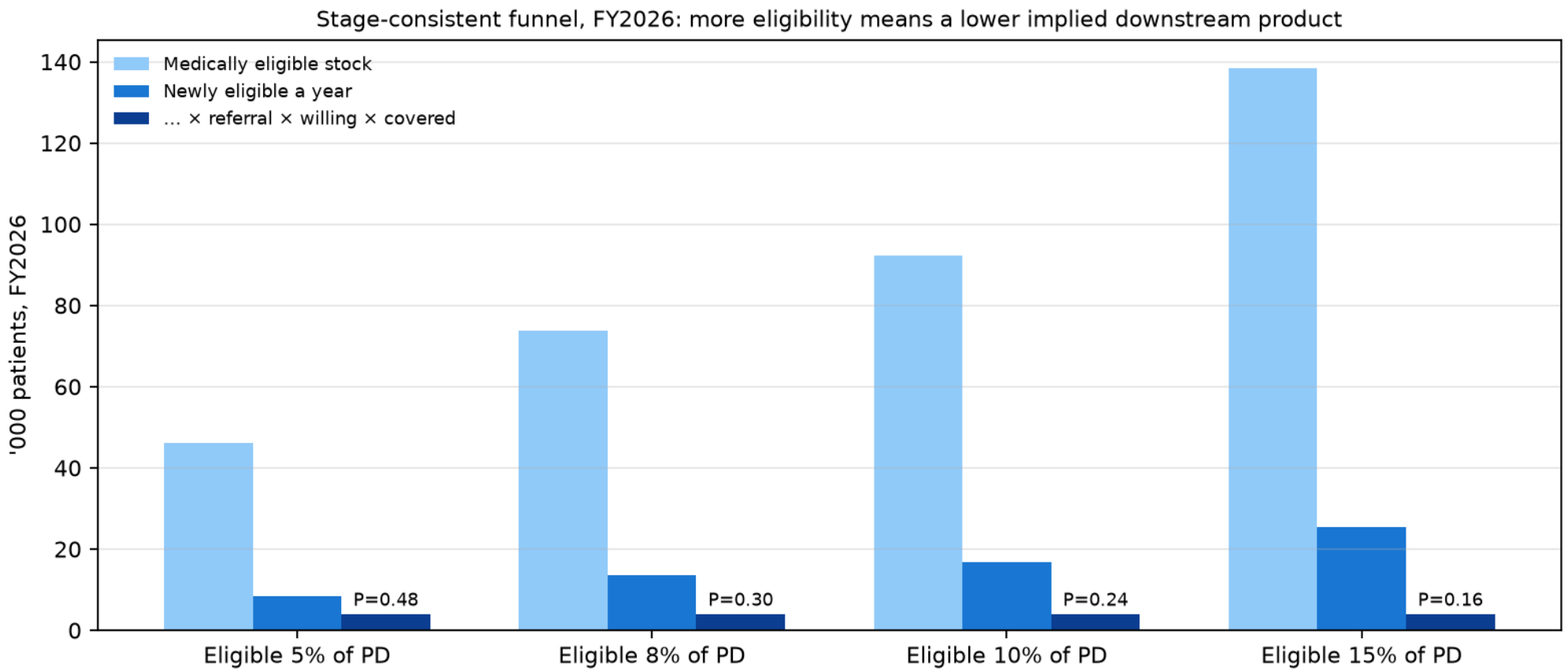
- **How the pool is built.** US diagnoses (90,000 a year [A1]) are spread by age using Hirsch incidence [A3] on Census 2024 population [A17]. They grow 2.6% a year (C1). Each cohort is followed through:
- survival: SSA 2023 life table [A4] × mortality ratio 1.52 [A5];
- dementia: 6.0% a year if diagnosed at 65 or older [A6], 4.3% a year if younger [A15];
- motor complications: 12% a year in years 2–5 and 22% a year after [A7, A15];
- the label's 4-year floor [B1].
- **Published screens alone.** Applying levodopa response 80.6% [A9], not adequately controlled 70% (★ S1) and fit 90% (★ S2) gives 4.3% of PD in FY2026. That share sets the age mix of the pool. The eligibility scenario then sets the level: published-screen counts are scaled uniformly by g = scenario ÷ 4.3%.
- **Relationship between the two measures.** Point prevalence s ≈ lifetime fraction (LF) × mean years eligible ÷ mean survival from diagnosis. In the cohort, an entrant stays eligible for 6.1 years on average, against 12.0 years of survival after diagnosis. So s ≈ LF × 0.51. The small gap to the scenario value (row 7 below) comes from incidence growth and age mix (C).
- **Olmsted floor.** In Olmsted County, 16% of incident PD developed fluctuations and 29% of those had DBS. That is about 4.8% ever-DBS [A8], so LF ≥ 4.8%. If ever-DBS = LF × P, then LF = 4.8% ÷ P (row 10).
- **What the Olmsted check does and does not show.** The model's ever-DBS share is 4.9% in every case, close to Olmsted's 4.8%. That supports the DBS calibration, but it cannot tell the eligibility cases apart, because P adjusts to offset eligibility.

Figure 2. Eligibility scenarios and implied downstream product

Row	Measure	5% (clinic studies)	8% (central ★)	10% (abstract)	15% (Aspen 150k)
3	Eligible, point prevalence of US PD	5.0%	8.0%	10.0%	15.0%
4	Scale on published screens g	1.17×	1.87×	2.33×	3.50×
5	Implied P = referral × willing × covered	0.48	0.30	0.24	0.16
6	Lifetime fraction of incident PD ever eligible	10.3%	16.5%	20.6%	30.9%
7	Check: LF × years eligible ÷ survival	5.3%	8.4%	10.5%	15.8%
8	Eligible stock, FY2026 ('000)	46.1	73.8	92.3	138.4
9	Newly eligible a year, FY2026 ('000)	8.4	13.5	16.9	25.3
10	Olmsted-implied LF = 4.8% ÷ P	10.0%	16.0%	20.0%	30.0%
11	P with DBS target 3.4k / 4.0k / 4.5k	0.48 / 0.56 / 0.64	0.30 / 0.35 / 0.40	0.24 / 0.28 / 0.32	0.16 / 0.19 / 0.21
13	DBS volume implied at judgment P = 0.30 ('000)	2.1	3.4	4.2	6.3
14	Plausibility of P	top of range	central	plausible	bottom of range

Source: pd_pool_model.xlsx, Scenarios rows 3–14 (row 11 from pd_pool_model.py, dbs_tests; target 4.0k = all 3,990 treated as under 75). Plausible range for P: 0.10–0.65, central 0.30 (★; referral 30–80%, willing 40–85%, covered 80–95%).

- **Why 8% is central (★).**
- It is the only case where the calibrated P (0.30) matches a stage-by-stage judgment (0.30).
- It lies between the measured clinic studies (Morgante 4.5% flexible, D-DATS 6.6% for any device therapy [B10]) and the 10–15% figures.
- The 10–15% figures are a survey abstract [A16] and Aspen's unsourced company belief [D5].
- **The 5% case is possible but demanding.**
- It needs about 80% referral × 65% willing × 92% covered.
- Aspen itself says only 10–15% of eligible patients are referred [D5]. That claim is unsourced, but it points the other way.
- **The 15% case needs P = 0.16.** That is consistent with poor referral (for example 30% × 60% × 90%), but it sits uneasily with Olmsted's finding that DBS was planned for 78% of untreated early-onset patients with fluctuations [A8].
- **Testing a higher DBS target** (4.5k, for outpatient implants) raises every P by about a third. The central case would then imply P = 0.40, still plausible.



Source: pd_pool_model.xlsx, Scenarios rows 5, 8, 9 and Flow sheets column H.

Figure 3. Central 8% case by age band, FY2026 ('000)

Age band	PD prevalent	Alive, no dementia, ≥4 yrs	+ motor complications	Eligible (central)	Eligible % of band PD	In funnel (× P)
<50	8.8	2.5	1.2	1.1	12.5%	0.3
50–59	39.5	14.9	9.0	8.5	21.5%	2.6
60–64	56.7	16.5	10.7	10.1	17.8%	3.0
65–69	92.9	37.8	22.4	21.2	22.9%	6.4
70–74	154.8	52.1	34.6	32.8	21.2%	9.8
75–79	197.0	0.0	0.0	0.0	0.0%	0.0
80+	373.0	0.0	0.0	0.0	0.0%	0.0
Total	**922.8**	**123.7**	**77.8**	**73.8**	**8.0%**	**22.1**

Source: pd_pool_model.xlsx, 'Age-band funnel' rows 3–10. 'In funnel' is the stock equivalent and is shown for scale only; the flow model applies P to inflow.

3. Calibration and checks

Figure 4. Checks

Check	Published	Model	Result
US PD prevalent 2020 ('000)	930.0 minimum [A19]	792.4	model 15% low
US PD prevalent 2030 ('000)	1,238.0 [A19]	1,021.5	model 17% low
DBS under 75, FY2019 ('000)	3.4 [A10, S5]	3.4	matched by construction
Ever-DBS share of incident PD	4.8% all ages [A8]	4.9% under 75	consistent
Bound B, kf = 100%, FY2045 vs funnel inflow ('000)	identity	6.6 vs 6.6	equal
xlsx flows vs Python, FY2001–45, all cases	—	max relative difference below 1e-12	pass

Source: pd_pool_model.xlsx, Checks and Summary; LibreOffice recalculation compared with pd_pool_model.json.

What the 3,990 covers [A10]. It is weighted NIS discharges in 2019 with a PD diagnosis and a DBS procedure code. That has three consequences: - **Inpatient only.** Outpatient implants are missed, so true volume is likely higher. - **All ages.** S5 assumes 85% are under 75; this is unverified. - **May not be one patient per discharge.** Staged second-side procedures could be counted twice. The calibration is therefore tested from 3.0k to 4.5k (Figure 2 row 11, Figure 7). The PD prevalence gap against Marras is inherited from the incidence inputs. Marras calls its own figures minimums.

4. Yearly flow, FY2026–2045

- **Pool.** Pool(t) = pool(t−1) × (1 − μ) + newly eligible × P − treated. The exit rate μ (death, dementia, ageing past 75) is about 16% a year.
- **Before launch.** DBS treats min(3.4k, pool) each year. Calibrating to FY2019 means DBS absorbs the whole funnel up to 2019. Inflow then grows 2.6% a year while DBS is held flat, so a backlog of willing, covered patients builds, reaching 2.8k by FY2028.
- **After launch (FY2029, ★ S7).**
- **Bound A**: DBS keeps its volume, and the one-time therapy treats 50% of the remaining pool each year (★ S6).
- **Bound B**: the one-time therapy takes over DBS's volume and also draws down the backlog at the same rate. B minus A therefore equals the DBS volume.
- **Tornado anomaly, fixed and explained.** Bound B now moves with DBS volume only because P is calibrated to that volume: more DBS implies a larger funnel. It no longer moves for any other reason.

Figure 5. Class demand, central 8% case ('000 patients; revenue $M at $20k)

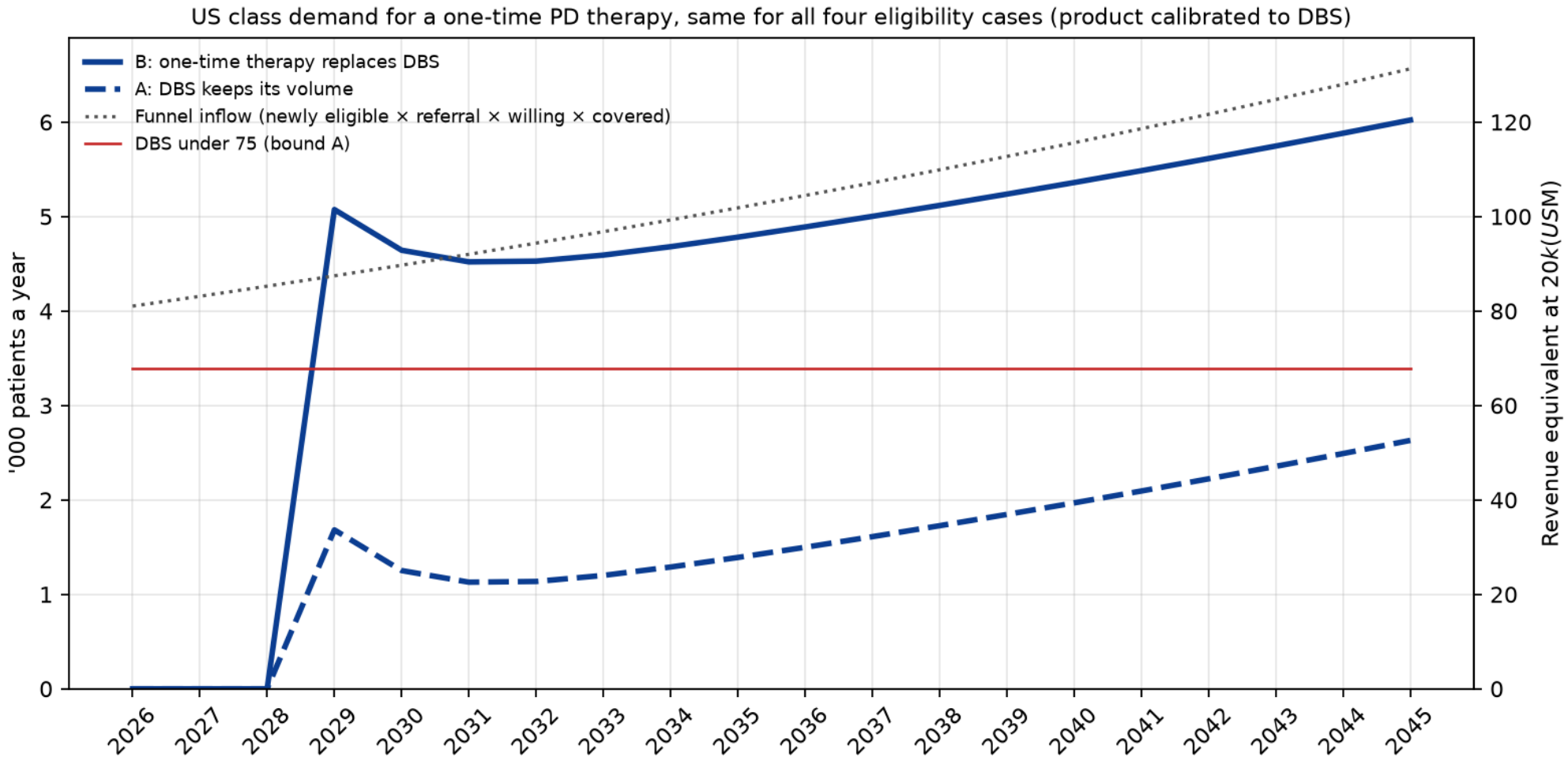
FY	Newly eligible	Funnel inflow (× P)	Deaths	Dementia	Aged past 75	Pool available	A: DBS	A: one-time	A: $M	B: one-time	B: $M	Backlog end, B
2026	13.5	4.1	1.9	2.8	7.0	5.3	3.4	0.0	0.0	0.0	0.0	1.9
2027	13.9	4.2	1.9	2.9	7.2	5.7	3.4	0.0	0.0	0.0	0.0	2.4
2028	14.2	4.3	1.9	3.0	7.4	6.2	3.4	0.0	0.0	0.0	0.0	2.8
2029	14.6	4.4	2.0	3.0	7.6	6.8	3.4	1.7	33.6	5.1	101.5	1.7
2030	15.0	4.5	2.0	3.1	7.8	5.9	3.4	1.3	25.0	4.6	92.9	1.3
2031	15.3	4.6	2.1	3.2	8.0	5.6	3.4	1.1	22.6	4.5	90.4	1.1
2032	15.7	4.7	2.2	3.3	8.2	5.7	3.4	1.1	22.7	4.5	90.6	1.1
2034	16.6	5.0	2.3	3.4	8.6	6.0	3.4	1.3	25.8	4.7	93.6	1.3
2035	17.0	5.1	2.3	3.5	8.8	6.2	3.4	1.4	27.8	4.8	95.6	1.4
2036	17.4	5.2	2.4	3.6	9.0	6.4	3.4	1.5	30.0	4.9	97.8	1.5
2038	18.3	5.5	2.5	3.8	9.5	6.8	3.4	1.7	34.5	5.1	102.4	1.7
2040	19.3	5.8	2.6	4.0	10.0	7.3	3.4	2.0	39.4	5.4	107.2	2.0
2042	20.3	6.1	2.8	4.2	10.5	7.8	3.4	2.2	44.5	5.6	112.3	2.2
2045	21.9	6.6	3.0	4.5	11.4	8.7	3.4	2.6	52.6	6.0	120.4	2.6

Source: pd_pool_model.xlsx, 'Flow 8%' rows 29–48 (columns C, H, E–G, N, J, K, M, P, R, Q). Exits are before treatment.

Figure 6. Class demand by eligibility case ('000 patients; $M at $20k)

Case	Implied P	Eligible FY2026	Newly eligible FY2026	Bound	FY2029	FY2030	FY2032	FY2035	FY2040	FY2045	$M FY2040
5%	0.48	46.1	8.4	A	1.7	1.3	1.1	1.4	2.0	2.6	39.4
5%	0.48	46.1	8.4	B	5.1	4.6	4.5	4.8	5.4	6.0	107.2
8%	0.30	73.8	13.5	A	1.7	1.3	1.1	1.4	2.0	2.6	39.4
8%	0.30	73.8	13.5	B	5.1	4.6	4.5	4.8	5.4	6.0	107.2
10%	0.24	92.3	16.9	A	1.7	1.3	1.1	1.4	2.0	2.6	39.4
10%	0.24	92.3	16.9	B	5.1	4.6	4.5	4.8	5.4	6.0	107.2
15%	0.16	138.4	25.3	A	1.7	1.3	1.1	1.4	2.0	2.6	39.4
15%	0.16	138.4	25.3	B	5.1	4.6	4.5	4.8	5.4	6.0	107.2

Source: pd_pool_model.xlsx, Summary rows 3–18 and Scenarios rows 5, 8, 9. Demand is identical across cases because P offsets eligibility.



Source: pd_pool_model.xlsx, 'Flow 8%' columns H, J, K, P (identical in the 5%, 10%, 15% sheets). Scenario, not guidance.

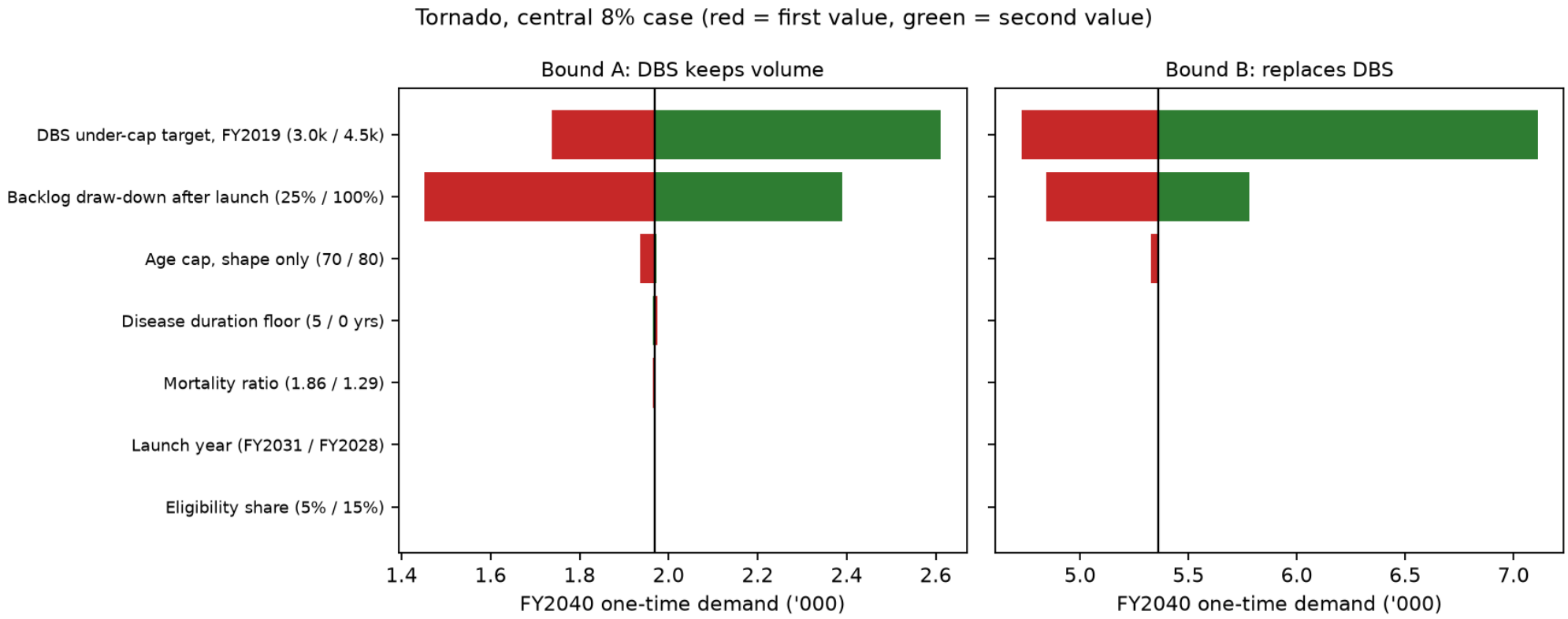
Convergence. When yearly uptake is 100% (kf = 1, the rerun), bound B equals the funnel inflow every year after launch: 6.6k against 6.6k in FY2045. Bound A equals the inflow minus DBS, 3.2k. At the base kf of 50%, bound B in FY2045 is 6.0k, or 92% of inflow. The gap is patients who leave the pool before they are treated.

5. Sensitivity

Figure 7. One-at-a-time sensitivity, central case: FY2040 one-time demand ('000)

Input (first / second value)	Bound A	Bound B	Implied P
Base	2.0	5.4	0.30
DBS under-cap target, FY2019 (3.0k / 4.5k)	1.7 / 2.6	4.7 / 7.1	0.26 / 0.40
Backlog draw-down after launch (25% / 100%)	1.5 / 2.4	4.8 / 5.8	0.30 / 0.30
Age cap, shape only (70 / 80)	1.9 / 2.0	5.3 / 5.4	0.27 / 0.31
Disease duration floor (5 / 0 yrs)	2.0 / 2.0	5.4 / 5.4	0.31 / 0.30
Mortality ratio (1.86 / 1.29)	2.0 / 2.0	5.4 / 5.4	0.32 / 0.28
Launch year (FY2031 / FY2028)	2.0 / 2.0	5.4 / 5.4	0.30 / 0.30
Eligibility share (5% / 15%)	2.0 / 2.0	5.4 / 5.4	0.48 / 0.16

Source: pd_pool_model.xlsx, Sensitivity. The age cap, mortality and duration floor are rerun through the cohort with the 8% eligibility level held fixed.



Source: pd_pool_model.xlsx, Sensitivity.

6. Judgment inputs (★)

- S8 Eligibility, point prevalence: 8% central; scenarios 5%, 10%, 15%.
- S1 / S2 Not adequately controlled 70% and medically fit 90%. These shape the age mix only; the level comes from S8.
- S3 Age cap 75 (the label has none; tested 70–80).
- S5 Share of DBS patients under 75: 85% (tested via a 3.0k–4.5k target).
- S6 Backlog draw-down after launch: 50% a year (25–100%).
- S7 Launch FY2029 (tested FY2028–31).
- S9–S11 and S4 Referral 55% (30–80%), willing 60% (40–85%), covered 90% (80–95%). Used only to judge whether the calibrated P is plausible.
- The uniform scale g assumes that higher eligibility has the same age and duration mix as the published-screen pool.

7. Why DBS-style eligibility understates a disease-modifying therapy's pool (qualitative; excluded from numbers)

- **Which patients qualify.** DBS treats symptoms, so it is offered late, once complications are no longer adequately controlled; for many it is a last resort. A therapy that slowed or reversed degeneration would make most sense early, before motor complications, which the label-style pool here excludes.
- **How willing they are.** The risk–benefit trade-off differs. Patients and neurologists may accept surgery earlier for a durable or disease-modifying benefit, so the 35–43% willingness to accept DBS would not carry over.
- **Why it is excluded.** None of the four partner therapies has shown disease modification, and exPDite-2 and REGENERATE-PD enrol patients with established motor fluctuations [D1, D2]. Any such upside belongs to a separate scenario with its own evidence.

8. Not verified or open

- No population-based study applies the label or EARLYSTIM criteria, so all four eligibility levels rest on clinic studies, surveys or company statements.
- The Olmsted figures (16%, 29%, 78%) come from an AAN abstract. 4.8% is our product of the two published figures (C).
- The share of DBS patients under 75, the outpatient share of implants, and double counting of staged procedures in the NIS are all unverified.
- Referral, willingness and coverage are separate judgments. Published figures map to them only loosely: PARADISE (device therapy 'not considered' for 26.6%), Aspen's unsourced 10–15% referral figure, and 58.7% implanted after evaluation [B10, D5, A21].
- DBS volume is held flat at the 2019 level throughout. Before 2019 the model's DBS is capped by the funnel, so it is lower than 3.4k.
- Dementia and motor complications are assumed independent within each age × duration cell. Modelled PD prevalence runs 15–17% below Marras.

Sources

- **D1** exPDite-2, NCT06944522: https://clinicaltrials.gov/study/NCT06944522
- **D2** REGENERATE-PD, NCT06285643: https://clinicaltrials.gov/study/NCT06285643
- **D3** ASPIRO, NCT06344026: https://clinicaltrials.gov/study/NCT06344026
- **D4** REPLACE, NCT07106021: https://clinicaltrials.gov/study/NCT07106021
- **D5** Aspen Neuroscience S-1, Oct 2026: https://www.sec.gov/Archives/edgar/data/2124080/000119312526419104/na-20261009.htm
- **B1** FDA P960009/S478 (Medtronic DBS indication), Feb 2025: https://www.accessdata.fda.gov/cdrh_docs/pdf/P960009S478B.pdf
- **B7** Schüpbach et al., EARLYSTIM, NEJM 2013: https://www.nejm.org/doi/full/10.1056/NEJMoa1205158
- **B10** Moes et al. 2023 review (Morgante, D-DATS, PARADISE, Malaty): https://pmc.ncbi.nlm.nih.gov/articles/PMC10645650/
- **A1** Parkinson's Foundation statistics: https://www.parkinson.org/understanding-parkinsons/statistics/prevalence-incidence

- **A3** Hirsch et al., Neuroepidemiology 2016: https://doi.org/10.1159/000445751
- **A4** SSA Period Life Table 2023: https://www.ssa.gov/oact/STATS/table4c6.html
- **A5** Macleod et al., Mov Disord 2014: https://doi.org/10.1002/mds.25898
- **A6** Williams-Gray et al., JNNP 2013: https://doi.org/10.1136/jnnp-2013-305277
- **A8** Stang et al., Rochester Epidemiology Project, Neurology 2021 suppl (abstract): https://doi.org/10.1212/WNL.96.15_supplement.4377
- **A9** Rizzo et al., Neurology 2016: https://doi.org/10.1212/WNL.0000000000002350
- **A10** Venkatraman et al., J Neurosurg 2024 (NIS): https://pmc.ncbi.nlm.nih.gov/articles/PMC10898494/
- **A11** DBS, developing world's perspective, MDCP 2023 (35%): https://pmc.ncbi.nlm.nih.gov/articles/PMC10715347/
- **A12** Dinkelbach et al., BMC Neurology 2017 (43.2%): https://doi.org/10.1186/s12883-017-0820-7
- **A15** Hely et al., Mov Disord 2005: https://doi.org/10.1002/mds.20324
- **A16** GMS 2012 abstract (10–15%): https://doi.org/10.3205/12dgnc047
- **A17** US Census Vintage 2024: https://www2.census.gov/programs-surveys/popest/datasets/2020-2024/national/asrh/nc-est2024-agesex-res.csv
- **A19** Marras et al., npj Parkinson's Disease 2018: https://doi.org/10.1038/s41531-018-0058-0
- **A21** Multidisciplinary DBS screening, MDCP 2018 (58.7%): https://pmc.ncbi.nlm.nih.gov/articles/PMC6183455/
- **A7** Ahlskog & Muenter, Mov Disord 2001 (motor fluctuations): https://doi.org/10.1002/mds.1090
- **P** $20k per procedure: instruction from David May.